Solvable groups and Nilpotent groups.

Def. 1.1. (1) The commutator of x and y : $[x,y] = x^{-1}y^{-1}xy$.

(2) The commutator group of G: $G^{der} = G' = [G,G] = \langle [x,y] \mid x,y \in G \rangle$

Remark. (1) $xy = yx$ iff $[x,y] = 1$.

(2) $g[x,y]g^{-1} = [gxg^{-1}, gyg^{-1}]$

(3) $G' \trianglelefteq G$, G/G' is Abelian (which is the motivation of the definition)

pf. (3) (sketched) $\forall\ \bar{x} \neq \bar{y} \in G/G'$. $[x,y] \in G' \Rightarrow [\bar{x},\bar{y}] = \bar{1} \Rightarrow \bar{x}\bar{y} = \bar{y}\bar{x}$.

Def 1.2. Solvable groups: a group G is called solvable iff there exists a chain: $\{1\} = G_0 \leq G_1 \leq \dots \leq G_r = G$

satisfying 1) $G_{i-1} \trianglelefteq G_i$; 2) G_i/G_{i-1} is Abelian.

Def 1.3. For any group G, define the following sequence:
$G^{(0)} = G$, $G^{(1)} = [G,G]$, $G^{(i+1)} = [G^{(i)}, G^{(i)}]$. is called the commutator series of G.

Prop. 1.4. A group is solvable iff $G^{(n)} = \{1\}$ for $n < \infty$.

pf: "$\Leftarrow$" is trivial;

"$\Rightarrow$": $\{1\} = H_0 \trianglelefteq H_1 \trianglelefteq \dots \trianglelefteq H_r = G$. ; G_i/G_{i-1} is Abelian.

thus $[H_i, H_i] \leq H_{i-1}$. (G/N Abelian $\Leftrightarrow$ $\ker\pi \supseteq [G,G]$).

thus. $[G,G] \leq H_{r-1}$

$G^{(2)} \leq [H_{r-1}, H_{r-1}] \leq H_{r-2} \Rightarrow \dots \Rightarrow G^{(r)} \leq H_0 = \{1\}$. □.

Def 1.5. Characteristic groups: G a group. $H \leq G$ is characteristic iff

for any automorphism σ of G, $\sigma(H) = H$. noted by $H \ \mathrm{char}\ G$

Prop. 1.6. (1) Characteristic groups are normal.

(2) All $G^{(i)}$ are characteristic groups of G.

(3) $H \leq G$ then $H^{(i)} \leq G^{(i)}$, thus G is solvable $\Rightarrow$ H is solvable.

(4) $\phi: G \to K$ surjective, homomorphism. then $\phi(G^{(i)}) = K^{(i)}$

thus G solvable $\Rightarrow$ K solvable.

(5) $N \trianglelefteq G$ and N, G/N are solvable, then G solvable.

pf: (trivial).

Def 1.7. (1) G a group. define the sequence

$G^0 = G$. ; $G^1 = [G,G]$; $G^{i+1} = [G, G^i]$ is called the lower central series of G.

(2) G is called nilpotent iff $G^c = \{1\}$ for $c \in \mathbb{N}$.

c : nilpotente class of G.

Remark. 1). $G^i \trianglelefteq G$. ; $G^{(i)} \leq G^i$.

2). let $N \trianglelefteq G$; $\pi : G \to G/N = H$ then $\pi(G^i) = H^i$.

3). Nilpotent groups are solvable.

Def 1.8. G group. define the sequence.

$Z_0(G) = 1$. $Z_1(G) = Z(G)$. $Z_2(G) = \pi^{-1}(Z(G/Z(G)))$.

$\cdots$ $Z_{i+1}(G) = \pi^{-1}(Z(G/Z_i(G)))$.

$$\begin{array}{ccc} G & \xrightarrow{\pi} & G/Z_i(G) \\ \cup & & \cup \\ Z_{i+1}(G) & \longrightarrow & Z(G/Z_i(G)) \end{array}$$

that is $Z_{i+1}(G)/Z_i(G) \cong Z(G/Z_i(G))$.

$\{1\} = Z_0(G) \trianglelefteq Z_1(G) \trianglelefteq \cdots$ is called the upper central series of G.

Thm 1.9. A group G in nilpotent iff $G = Z_c(G)$ for some $c \in \mathbb{N}$.

pf: Induction on the minimal c such that either $G^c = \{1\}$ or $Z_c(G) = Z(G)$.

$c = 1$. $Z(G) = G \iff G$ Abelian $\iff [G,G] = G' = \{1\}$

for all $< c$ the proposition is correct.

Let $\pi : G \to G/Z(G) := \overline{G}$.

we will prove: $G^{\overset{c}{\leftarrow}} = \{1\}$ $\quad\quad$ $Z_c(G) = G$

$\Updownarrow$ (1) $\quad\quad$ $\Updownarrow$ (2)

$(\overline{G})^{\overset{c-1}{\leftarrow}} = \{1\}$ $\overset{\text{inductive hypothesis}}{\Longleftrightarrow}$ $Z_{c-1}(\overline{G}) = \overline{G}$

(1). $G^c = \{1\} \iff [G, G^{c-1}] = \{1\} \iff G^{c-1} \leq Z(G) \iff (\overline{G})^{c-1} = \{1\}$.

(2).

$$\begin{array}{ccccccc} G & \supseteq & Z_1(G) & \hookrightarrow & Z_2(G) & \hookrightarrow & Z_3(G) \\ \downarrow \pi_1 & & & & \downarrow \pi_1 & & \downarrow \pi_2 \\ G/Z(G) = \overline{G} & & \supseteq & & Z_1(\overline{G}) & \hookrightarrow & Z_2(\overline{G}) \\ \downarrow \overline{\pi}_1 & & & & & & \downarrow \overline{\pi}_1 \\ G/Z_2(G) = \overline{G}/Z_1(\overline{G}) & & & & & & Z(\overline{G}/Z_1(\overline{G})) \end{array}$$

(π_2 : $G \to G/Z_2(G)$)

thus $\dfrac{G}{Z_2(G)} \cong \dfrac{G/Z_1(G)}{Z_2(G)/Z_1(G)} = \dfrac{\overline{G}}{Z_1(\overline{G})}$. (*)

$\Rightarrow Z(G/Z_2(G)) \cong Z(\overline{G}/Z_1(\overline{G}))$. (***)

At the same time, $Z_2(\overline{G}) = \overline{\pi}_1^{-1}(Z(\overline{G}/Z_1(\overline{G})))$ (**)

$\Rightarrow Z_3(G) = \pi_2^{-1}(Z(G/Z_2(G))) \overset{*}{=} \pi_1^{-1}(Z(\overline{G}/Z_1(\overline{G}))) \overset{**}{=} \pi_1^{-1}(Z_2(\overline{G}))$.

(π_1^{-1} : $\overline{G} \to G$)

More generally: $Z_n(G) = \pi_1^{-1}\left(Z_{n-1}(\bar{G})\right)$.

$\Rightarrow$ thus: $Z_c(G) = G \iff Z_{c-1}(\bar{G}) = \bar{G}$. $\square$

Remark. More precisely, we have $G^{c-i} \le Z_i(G)$.

Since by inductive hypothesis, $\bar{G}^{c-i} \le Z_{i-1}(\bar{G})$

$\Rightarrow G^{c-i} \le \pi_1^{-1}(\bar{G}^{c-i}) \le \pi_1^{-1}\left(Z_{i-1}(\bar{G})\right) = Z_i(G)$

Thm 1.10. (Structure thm of nilpotent groups). let $|G| = n = p_1^{\alpha_1} \cdots p_r^{\alpha_r}$ $P_i \in Syl_{p_i}(G)$

Then TFAE: (1) G is nilpotent;

(2) $H \subsetneq G \Rightarrow H \subsetneq N_G(H)$.

(3) All sylow subgroups P_i are normal

(4) $G \cong P_1 \times P_2 \times \cdots \times P_r$.

Lemma 1.11. All p-groups are nilpotent.

pf: $|P| = p^\alpha$, $(\alpha \ge 1)$. then $p^\alpha = |Z(P)| + \sum_{i=1}^{r} |P : C_P(h_i)|$

↳ nontrivial p-power

$\Rightarrow p \mid |Z(P)|$, $\{e\} \le Z(P)$ thus $Z(P)$ is nontrivial $\square$.

Lemma 1.12. Let P be a p-group.

(1) $Z(P) \neq \{e\}$.

(2) If $H \trianglelefteq P$. H nontrivial. then $H \cap Z(P) \neq \{e\}$

(3) If $H \subsetneq P$. then $H \subsetneq N_P(H)$.

pf: (1) ✓; (2) Consider the conjugation action of P on H:

then $H = \coprod_i P/Stab_P(a_i)$ where $Stab_P(a_i) = P \iff a_i \in Z(P) \cap H$

$\Rightarrow 0 \equiv |H| \equiv \sum_{i=1}^{r} |P|/|Stab_P(a_i)| = |Z(P) \cap H| \pmod p$

Since H is nontrivial, thus $H \cap Z(P) \neq \{e\}$

(3) ① $Z(P) \not\subseteq H$; $Z(P) \subseteq P \Rightarrow H \subsetneq N_P(H)$.

② $Z(P) \subseteq H$; consider $\bar{H} = H/Z(P)$. $\bar{P} = P/Z(P)$. By induction,

$\bar{H} \subsetneq N_{\bar{P}}(\bar{H})$. thus $H \subsetneq N_P(H)$. $\square$.

Back to the proof:

(1) $\Rightarrow$ (2) ① $Z(G) \not\subseteq H$, ...

② $Z(G) \le H$ consider $\bar{H} = H/Z(P)$, $\bar{P} = P/Z(P)$ ...

(2) $\Rightarrow$ (3) Note that $N_G(N_G(P_i)) = N_G(P_i)$. ($N_G(P_i) \trianglelefteq N_G(N_G(P_i))$; $N_G(N_G(P_i)) \subseteq N_G(P_i)$).

if $N_G(P_i) \subsetneq G \Rightarrow N_G(P_i) \subsetneq N_G(N_G(P_i)) = N_G(P_i)$ contradiction.

(3) $\Rightarrow$ (4) $P_i \cap P_j = \{e\}$. $(i \neq j)$. P_i (normal) $\Rightarrow G \cong P_1 \times \cdots \times P_r$.

(4) $\Rightarrow$ (1): P_i is nilpotent. then $G = \prod_{i=1}^{n} P_i$

$\Rightarrow Z^c(G) = \prod_{i=1}^{n} Z^c(P_i)$. take $c = \max\{c_1, \ldots, c_n\} \Rightarrow Z^{(c)}(G) = G$ $\square$

A general and brief introduction of the ring theory.

Def 2.1. (1). R is a ring iff (1) $(R,+)$ is an Abelian group

(2) $(R,\cdot)$ is a monoid.

(3). $1 \neq 0$

(4) for $a,b,c \in R$. $a(b+c) = a\cdot b + a\cdot c$; $a\cdot(b+c) = ab + a\cdot c$.

(2) R is a division ring iff $\forall x \in R$, $\exists\, x^{-1} \in R$. s.t. $x \cdot x^{-1} = x^{-1} \cdot x = e$

(3) R is a field iff R is a commutative division ring.

(4) R is an integral domain iff R is a commutative ring with no zero-divison.

In the following lecture, we just take R commutative.

Remark. A finite integral domain R is a field

pf: (sketched) Define $\phi_a : (R,+) \longrightarrow (R,+)$

$x \longmapsto ax$

Def 2.2 A left ideal is a subset $I \subseteq R$ iff (1) $a-b \in I$. $\forall a,b \in I$.

(2) $xa \in I$. $\forall x \in R$. $a \in I$.

Notation: define the ideal generated by $a_1, \cdots, a_s$: $(a_1, \cdots, a_s) = \{ \sum_{i=1}^{s} x_i a_i \mid x_i \in R \} \subseteq R$

Def 2.3. R ring, I ideal then the quotient ring: $R/I := \{ x+I \mid x \in R \}$

with operations: $x+I + y+I = (x+y)+I$. $(x+I)(y+I) = (xy)+I$

Examples: $\mathbb{R}[x]/(x^2+1) \cong \mathbb{C}$.

Thm 2.4. (1) $\phi : R \to S$. surjective, homo~ $\Rightarrow$ $S \cong R/\ker\phi$.

(2) $I \subseteq J$ ideal of R then $(R/I)/(J/I) \cong R/J$.

Def 2.5. The fraction field of the integral domain R

$\mathrm{Frac}(R) := \{(a,b) \in R \times (R \setminus \{0\})\} / \sim$.

where $(a,b) \sim (c,d)$ iff $ad = cb$. hence $\mathrm{Frac}(R)$ is a field

Examples: $\mathrm{Frac}(\mathbb{Z}) = \mathbb{Q}$. $\mathrm{Frac}(k[x]) = k(x)$

Thm 2.6. (Chinese remainder thm)

Let $I_1, \dots, I_k$ be ideals of R. $I_1, \dots, I_k$ are pairwise comaximal

then (1) $\phi: R \longrightarrow R/I_1 \times \dots \times R/I_k$ is surjective;

(2) $I_1 \cap \dots \cap I_k = I_1 \dots I_k$

(3) $R/I_1 \dots I_k \cong R/I_1 \cap \dots \cap I_k \cong \frac{R}{I_1} \times \dots \times \frac{R}{I_k}$

pf (skipped).

Corollary 2.7. $n_1, \dots, n_r$ are pair-wise coprime. then $\frac{\mathbb{Z}}{n_1\mathbb{Z}} \times \dots \times \frac{\mathbb{Z}}{n_r\mathbb{Z}} \cong \frac{\mathbb{Z}}{n_1 n_2 \dots n_r \mathbb{Z}}$

Def 2.8. Some special ideals.

(1). $m \subseteq R$ is called maximal if $m \neq R$ and the only ideals containing m are m & R

(2) A proper ideal $\mathfrak{p} \subsetneq R$ is called a prime ideal if

$ab \in \mathfrak{p} \Rightarrow a \in \mathfrak{p}$ or $b \in \mathfrak{p}$

Prop. 2.9. (1) $m \subseteq R$ is maximal iff R/m is a field.

(2) $\mathfrak{p} \subseteq R$ is a prime ideal iff $R/\mathfrak{p}$ is an integral domain

(3) maximal ideal $\Rightarrow$ prime ideal.

pf: consider the $\phi: m \to R/m$ & the lattice isomorphism thm. (sketched).

Remark. Prop. 10.4.5 is a good exercise.

Def. 2.9. Let R be an integral domain.

(1). R is said to be a UFD. if $\forall r \in R$. $\exists!$ $p_1, \dots, p_n$ irreducible (under the sense of association), s.t. $r = p_1 p_2 \dots p_n$.

(2) R is said to be a PID if every ideal is principle.

(3) R is said to be a ED if there exists $Nm: R \to \mathbb{Z}^+ \cup \{0\}$. s.t.

① $Nm(0) = 0$

② $\forall a, b \in R$. with $b \neq 0$. $\exists q \in R$. $r \in R$. s.t. $a = bq + r$. $0 \leq Nm(r) < Nm(b)$.

Some useful & basic properties.

Prop. 2.10. (1) R is UFD. then prime elements $\Leftrightarrow$ irreducible elements.

(2) R is PID then maximal ideal $\Leftrightarrow$ prime ideal.

(3) F is a field then $F[X]$ is an ED

Thm. 2.11. ED $\overset{(1)}{\Rightarrow}$ PID $\overset{(2)}{\Rightarrow}$ UFD.

pf: (1) $\Rightarrow$ (2) Consider $b = aq + r$. $0 \le Nm(r) < Nm(a)$.
(sketched)

(2) $\Rightarrow$ (3) Consider $r \in R$. $r = a_1 b_1 = a_1 a_2 b_2 = a_1 a_2 a_3 b_3 = \dots$

$\Rightarrow (r) \subseteq (b_1) \subseteq \dots$

Then take $I = \bigcup_{n \ge 0} (b_n) : \overset{\text{PID}}{=\!=} (b)$

Uniqueness: $r = p_1 \cdots p_n = q_1 \cdots q_m$ and use induction $\square$.

Thm.2.12. R is an integral domain, then R is a UFD $\Leftrightarrow$ R[x] is a UFD.

pf (sketched): $\Rightarrow$ Using Gauss Lemma: take $F = \text{Frac}(R)$. $p(x) \in R[x]$ is reducible in $F[x]$.
then $p(x)$ is reducible in $R[x]$ $\square$.

A useful lemma2.13(in the Field theory): F be a field. $f(x)$ is irreducible iff $F[x]/(f(x))$ is a field.

pf: $f(x)$ irreducible $\overset{\text{UFD}}{\Leftrightarrow}$ $f(x)$ prime elements
$\overset{\text{PID}}{\Leftrightarrow}$ $(f(x))$ maximal in $F[x]$.
$\Leftrightarrow$ $F[x] / (f(x))$ is a field $\square$.

Modules & Classification of finitely generated modules over PID

Def 3.1 ① Let R be a ring. A left R-module is an abelian group M equipped with an R-action on M:

$$R \times M \longrightarrow M$$
$$(a, m) \longmapsto a \cdot m.$$

satisfying (1). $1_R \cdot m = m$

(2) $(r+s)\cdot m = r\cdot m + s\cdot m$

(3) $r\cdot(m+n) = r\cdot m + r\cdot n$

(4) $(r\cdot s)m = r\cdot(s\cdot m)$.

② An R-submodule $N \subseteq M$ is an abelian subgroup of M.
s.t. $\forall r \in R$, $r \cdot N \subseteq N$.

Example. 1). $R = F$ a field $\Rightarrow$ F-vector spaces.

2). free modules $R^{\oplus n} = \{(a_1, \dots, a_n) \mid a_i \in R\}$.

$(a_1, a_2, \dots, a_n) + (b_1, \dots, b_n) = (a_1+b_1, \dots, a_n+b_n)$

$r(a_1, \dots, a_n) = (ra_1, \dots, ra_n)$

3) A $\mathbb{Z}$-module is an abelian group G. $\begin{cases} n\cdot g = g + \dots + g \\ (-n)\cdot g = -n\cdot g \end{cases}$

4). a left R-submodule of R $\Longleftrightarrow$ a left ideal of R

Def 3.2. ① Direct product: M_i $(i \in I)$ are left R-modules,

then $\prod_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i\}$

② Direct sum: $\bigoplus_{i\in I} M_i = \{(m_i)_{i\in I} \mid m_i \in M_i$, all but finitely many m_i's are zero$\}$.

Def 3.3. ① R-module homomorphism: (skip)
isomorphism.

② Definition of $\ker\phi$ & $\mathrm{Im}\,\phi$

③ quotient R-module: $M/N := \{m+N \mid m \in M\}$.

④ M is called irreducible iff the only submodule of M is 0 and M.

Remark. When R is a commutative ring.

for $\phi \in \mathrm{Hom}_R(M,N)$, $a \in R$. define $a\cdot\phi \in \mathrm{Hom}_R(M,N)$

by $(a\cdot\phi)(m) := a\cdot\phi(m) = \phi(am)$.

Hence $\mathrm{Hom}_R(M,N)$ is equipped with the structure of R-module.

$\left((a\cdot\phi)(bm) = (b\cdot a\cdot\phi)(m) = (ba)\phi(m) = (ab)\phi(m)\right)$.

Def 3.4. M a left R-module, $X \subseteq M$ a subset

Define $RX := \{a_1x_1 + \dots + a_nx_n \mid a_i \in R,\ x_i \in X\}$

(1) If $|X| < \infty$, then $N = RX$ is finitely generated by X.

(2) If $\exists\, m \in M$ s.t. $N = Rm$, then $N \subseteq M$ is called cyclic.

(3) M is free of rank n iff $\exists\, x_1, \dots, x_n \in M$. $\forall\, m \in M$, $\exists!\ (a_1, \dots, a_n) \in R^{\oplus n}$ s.t. $m = a_1x_1 + \dots + a_nx_n$

Next we will prove the main theorems in this lecture.

Thm 3.5. Let R be a PID. let N be a free R-module of rank n. and L a submodule of N.

Then (1) L is free of rank t ($t \le n$).

(2) There exists a basis $y_1, \dots, y_n$ of N so that $a_1y_1, \dots, a_ty_t$ is a basis of L and $a_1, \dots, a_t \in R \setminus \{0\}$. s.t. $a_1 \mid a_2$, $a_2 \mid a_3$, $\dots$, $a_{t-1} \mid a_t$.

Lemma 3.6. Let R be an integral domain. and let N be a free R-module of n. Then any $(n+1)$ elements $x_1, \dots, x_{n+1} \in N$ are dependent linearly.

pf: (sketched) Consider the fraction field of R. □

proof of Thm 3.5. Consider the isomorphism $N \cong R^{\oplus n}$.

Consider the set $\{\phi(L) \mid \text{for all } \phi \in \mathrm{Hom}_R(N, R)\} = S$

It contains a maximal ideal (a_1) where $\phi_1 : N \to R$ with $\phi_1(y_0) = a_1$. ($\exists\, y_0 \in R$)

(Since R is a PID, hence Noetherian ring). (this proof is easy). $I_1 \subsetneq I_2 \subsetneq \dots \subsetneq \dots$

consider $I = \bigcup_{n \ge 1} I_n$.

Then for any $\phi' \in \mathrm{Hom}_R(N, R)$ we have $a_1 \mid \phi'(y_0)$.

(In fact, $d = \gcd(a_1, \phi'(y_0)) \Rightarrow d = (r_1\phi + r_2'\phi')(y_0) \Rightarrow (d) \in S$ contradicting with maximality.)

$(d) \supsetneq (a_1)$

Now we have $\pi_i(y_0) = a_1 b_i$ $(b_i \in R) \Rightarrow y_0 = (a_1b_1, \dots, a_1b_n)$

$\Rightarrow a_1\phi_0(b_1, \dots, b_n) = \phi(y_0) = a_1$. let $y_1 := (b_1, \dots, b_n) \in N$.

$\Rightarrow \phi_0(y_1) = 1$.

Claim: (i) $N \cong Ry_1 \oplus \ker\phi$

(ii) $L = Ra_1y_1 \oplus (L \cap \ker\phi)$.

(i). $\forall\, x \in N$, we have $x = \underbrace{\phi(x)\cdot y_1}_{\in Ry_1} + \underbrace{x - \phi(x)\cdot y_1}_{\in \ker\phi}$.

$Ry_1 \cap \ker\phi = \{ry_1 \mid r\phi(y_1) = r = 0\} = \{0\}$

$\Rightarrow N \cong Ry_1 \oplus \ker\phi$.

(ii). $\forall\, x \in L$, we have $x = \underbrace{\phi(x)\cdot y_1}_{\phi(x) \in (a_1)} + \underbrace{(x - \phi(x)\cdot y_1)}_{\in L \cap \ker\phi}$.

Back to the proof: $L = Ra_1y_1 \oplus (L \cap \ker\phi) = Ra_1y_1 \oplus Ra_2y_2 \oplus \dots \oplus Ra_{n-1}y_{n-1}$ (by Lemma 3.6),

thus we prove (1)

$\dots\ m > n$

(3) If $M = R/(q^n)$. then $p^mM / p^{m+1}M = 0$.

(2). $L = Ra_1y_1 \oplus Ra_2y_2 \oplus \dots \oplus Ra_ty_t \subseteq N = Ry_1 \oplus \dots \oplus Ry_n$ (by (1) & # induction)

Suppose $a_1 \nmid a_2$. take $d = \gcd(a_1, a_2) = a_1r_1 + a_2r_2$ $(r_1, r_2 \in R)$

and homomorphism: ϕ': $N \longrightarrow R$

$y_1 \mapsto r_1$

$y_2 \mapsto r_2$

$y_i \mapsto 0$

then $\phi'(a_1y_1 + a_2y_2) = d\phi$ ♭ then $a_1 \nmid \phi'(a_1y_1 + a_2y_2)$ contradiction # with the construction in (1). So $a_1 | a_2$ similarly, we have $a_1 | a_2, \dots, a_{t-1} | a_t$ □.

Thm. 3.7. (Foundamental thm of finitely generated modules over PIDs).

Let R be a PID and L a finitely generated R-module.

Then $L \cong R^{\oplus r} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$. $(a_i | a_{i+1})$, moreover, it is unique

Pf: As L is finitely generated: $\exists x_i \in L$. ϕ: surjective, homomorphism

$\phi: R^{\oplus n} \longrightarrow\!\!\!\!\rightarrow M$.

$(a_1, \dots, a_n) \longmapsto a_1x_1 + \dots + a_nx_n$

by the isomorphism theory of module: $M \cong R^{\oplus n}/\ker\phi$

$\exists$ basis $y_1, \dots, y_n \in R^{\oplus n}$. and $a_1, \dots, a_m \in R$. s.t. $R^{\oplus n} = Ry_1 \oplus \dots \oplus Ry_n$.

& $\ker\phi = Ra_1y_1 \oplus \dots \oplus Ra_my_m$

$\Rightarrow M \cong \dfrac{Ry_1 \oplus \dots \oplus Ry_n}{Ra_1y_1 \oplus \dots \oplus Ra_my_m} \cong R^{\oplus(n-m)} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$

(the second isomorphism can be proved easily:

define φ: $Ry_1 \oplus \dots \oplus Ry_n \longrightarrow R^{\oplus(n-m)} \oplus R/(a_1) \oplus \dots \oplus R/(a_m)$.

$x \in Ry_i$ $(1 \le i \le m) \longmapsto \pi_i(x) + (a_i)$.

$x \in Ry_i$ $(m+1 \le i \le n) \longmapsto \pi_i(x)$

$\Rightarrow \ker\varphi = Ra_1y_1 \oplus \dots \oplus Ra_my_m$)

Next we will prove the hardest part: uniqueness.

by Chinese remainder theorem:

$M = R^{\oplus r} \oplus R/(p_1^{\alpha_{1,1}}) \oplus \dots \oplus R/(p_1^{\alpha_{1,s_1}}) \oplus R/(p_2^{\alpha_{2,1}}) \oplus \dots$

where $a_1 = p_1^{\min\{\alpha_{1,1}, \dots, \alpha_{1,s_1}\}} p_2^{\min\{\alpha_{2,1}, \dots, \alpha_{2,s_2}\}} \dots$

...

We want to show that: r, p_i, $\alpha_{i,j}$ can be determined by M itself!

Lemma 3.8. p, q are prime elements such that $(p) \neq (q)$.

then (1) If $M = R$. then $p^mM/p^{m+1}M \cong R/(p)$

(2) If $M = R/(p^n)$, then $p^mM/p^{m+1}M \cong \begin{cases} R/(p) & \text{if } m < n \\ 0 & \text{if } m > n \end{cases}$

(3) If $M = R/(q^n)$. then $p^mM/p^{m+1}M = 0$.

Back to the proof. we say that:

$$\dim_{R/(p_1)} (M/p_1^{k}M) = r + \#\{\alpha_{1,1}, \cdots, \alpha_{1,s_1}\}$$

$$\dim_{R/(p_1)} (pM/p_1^{2}M) = r + \#\{\alpha_{1,j} \geq 2\}.$$

...

Let p_0 varying all prime elements in R and all $m \in \mathbb{Z}_{\geq 0}$ we can determine all the numbers $r, p_1, \cdots p_r, \alpha_{1,1}, \cdots$ $\square$